# The Direct and Indirect Effects of Genetics and Education

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# The Direct and Indirect Effects of Genetics and Education

Senan Hogan-Hennessy, 14 November 2024.

**Notes for LWIPS:**

This project integrates data, methods from other fields to labour econ topics.

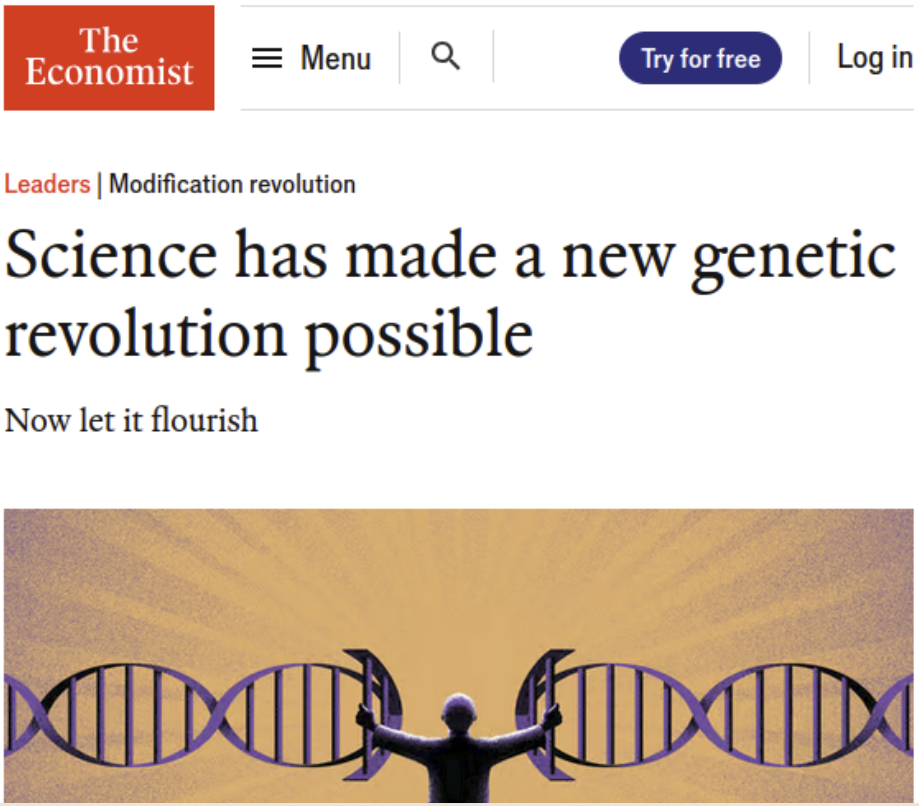
1. New genetic measures (from genetics)
2. An explicit mediation analysis (methods from statistics/epidemiology)
3. Aims to use these in way that is credible in labour econ, natural experiment setting.

Appreciate comments on what is confusing + unclear from an econ audience, given these unconventional approaches.

**Note:** Explains an overlapping research design approach for causal effects using UK Biobank data; causal results with these data are not ready here.

# Introduction

"**The difference of natural talents in different men**, is, in reality, much less than we are aware of [. . . ]. The difference between the most dissimilar characters, between a philosopher and a common street porter, seems to arise **not so much from nature, as from habit, custom, and education**." *A. Smith (1776), The Wealth of Nations, Volume 1 p. 16.*




## Nature vs Nurture

1. Genetics measure nature (initial endowment)
2. Find genes associated with outcomes
3. Deeper understanding of social outcomes, from genetics.

# Introduction

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## Nature(?) vs Nurture

1. Genetics measure one component of nature (at most. . . )
2. Genes associated with outcomes are not the entire story
3. Genetics gives new findings, **with caveats.**

INTERDISCIPLINARY RESEARCH

**Economics and genetics meet in uneasy union**

*Use of population-genetic data to predict economic success sparks war of words.*

BY EWEN CALLAWAY

"The invalid assumption that correlation implies cause is probably among the two or three most serious and common errors of human reasoning." Evolutionary biologist Stephen Jay Gould was referring to purported links between genetics and an individual's intelligence when he made this familiar complaint in his 1981 book *The Mismeasure of Man*.

Fast-forward three decades, and leading geneticists and anthropologists are levelling a similar charge at economics researchers who claim that a country's genetic diversity can predict the success of its economy. To critics, the economists' paper seems to suggest that a country's poverty could be the result of its citizens' genetic make-up, and the paper is attracting charges of genetic determinism, and even racism. But the economists say that they have been misunderstood, and are merely using genetics as a proxy for other factors that can drive an economy, such as history and culture. The debate holds cautionary lessons for a nascent field that blends genetics with economics, sometimes called genoeconomics. The work could have real-world pay-offs, such as helping policy-makers "reduce barriers to the flows of ideas and innovations across populations", says Enrico Spolaore, an economist at Tufts University near Boston, Massachusetts, who has also used global genetic-diversity data in his research.

But the economists at the forefront of this field clearly need to be prepared for harsh scrutiny of their techniques and conclusions. At the centre of the storm is a 107-page paper by Oded Galor of Brown University in Providence, Rhode Island, and Quamrul Ashraf of Williams College in Williamstown, Massachusetts[1]. It has been peer-reviewed by economists and biologists, and will soon appear in *American Economic Review*, one of the most prestigious economics journals.

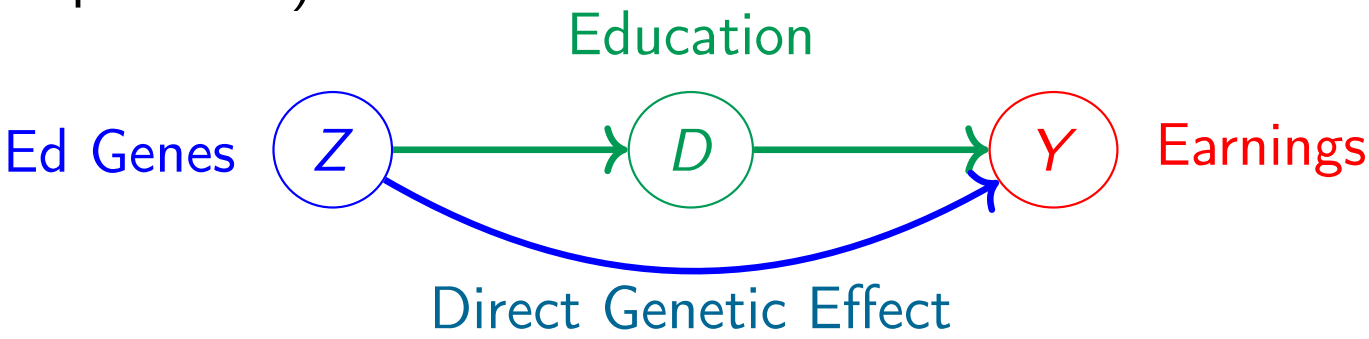
The paper argues that there are strong links between estimates of genetic diversity for 145 countries and per capita incomes, even

# Introduction

**How do genes associated with education affect labour outcomes?**

1. Causal design for random variation in a genetic measure for education.

2. Decompose mechanisms by a mediation analysis (with an overlapping natural experiment)




3. Finding that genes associated with education do/not affect outcomes **directly or indirectly,** inferring they do/not measure intelligence, aptitude, or other factors.

# Introduction

**How do genes associated with education affect labour outcomes?**
Write more here about the plan, and perhaps even why it matters.

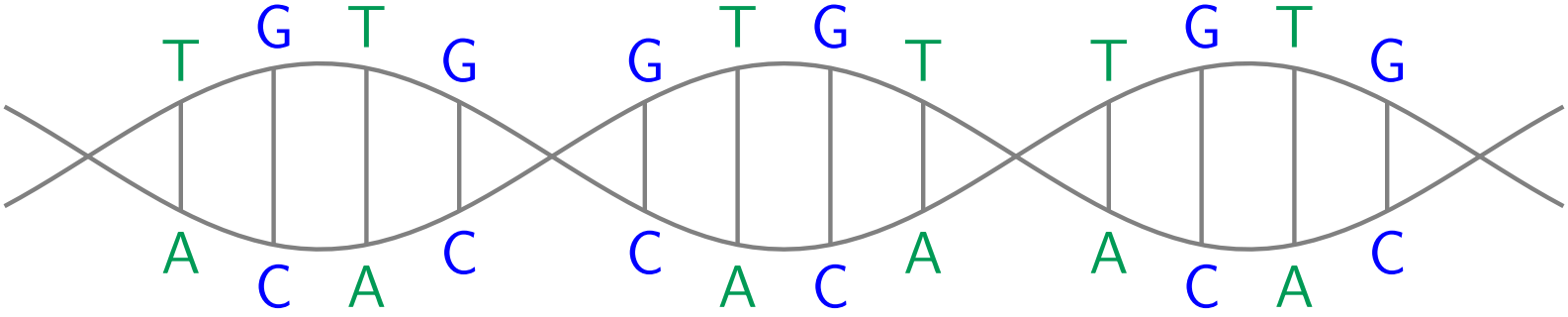
## Introduction

1. **Genetics & Education**
   Data: Health & Retirement Study
2. **Genetic Effects (Associations)**
   Mendelian Independent Assortment
3. **Direct and Indirect Effects**
   Mediation of Ed PGI Through Education

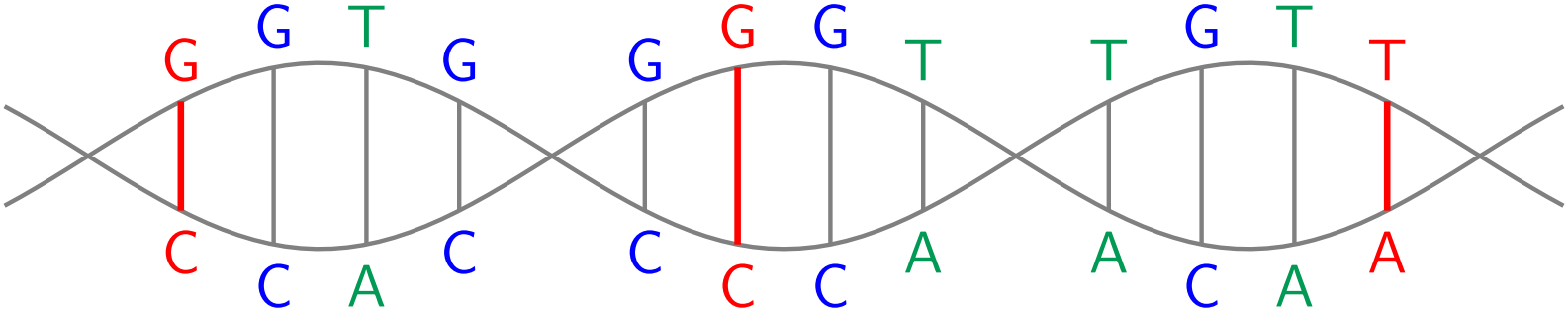
# Genetics: a Primer

- The human genome is a 3 billion-long sequence of base pairs
- $\approx 0.1\%$ these pairs differ between two average people

Person 1's DNA:




Person 2's DNA:



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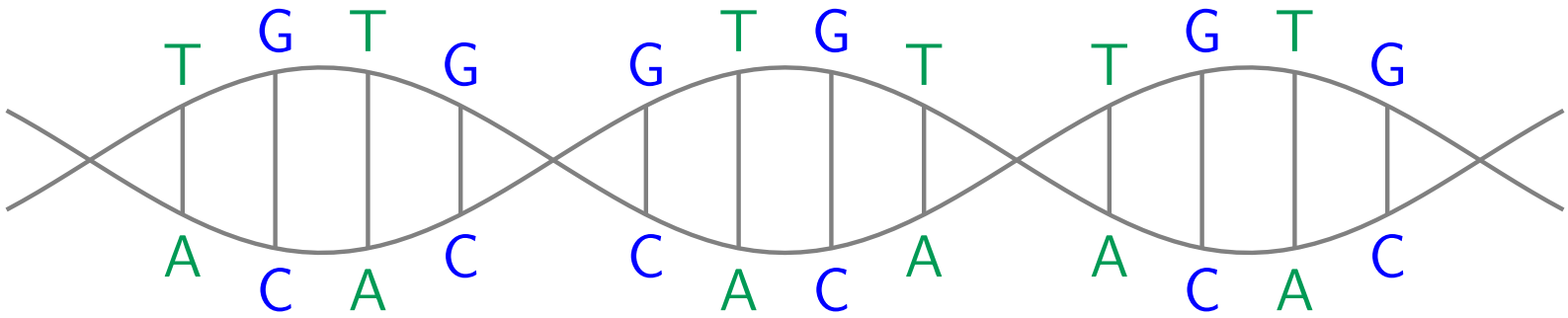


Gene:

Protein: $X$

Outcome

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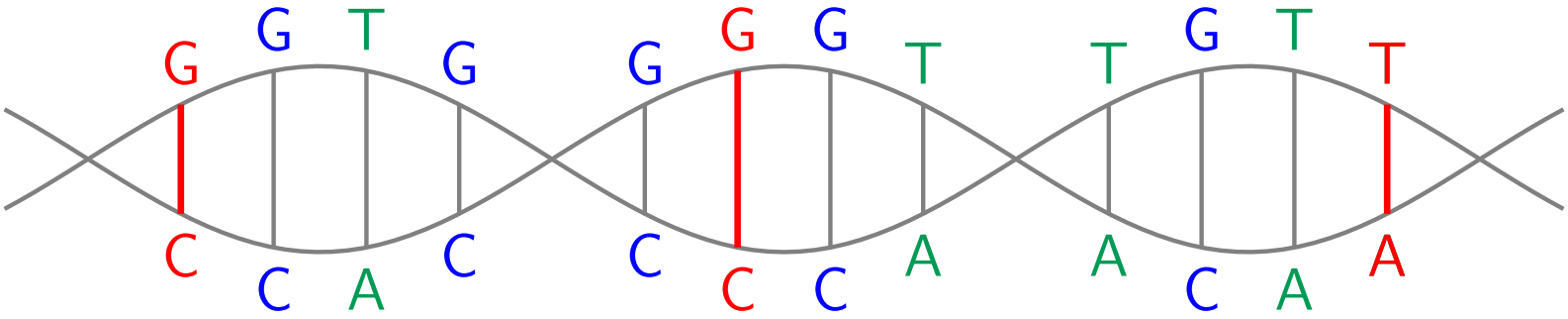


Gene:

Protein: $X$

Different outcome.

Moderated by external factors

*"gene–environment interactions."*

# Polygenic Index & Education (Ed PGI)

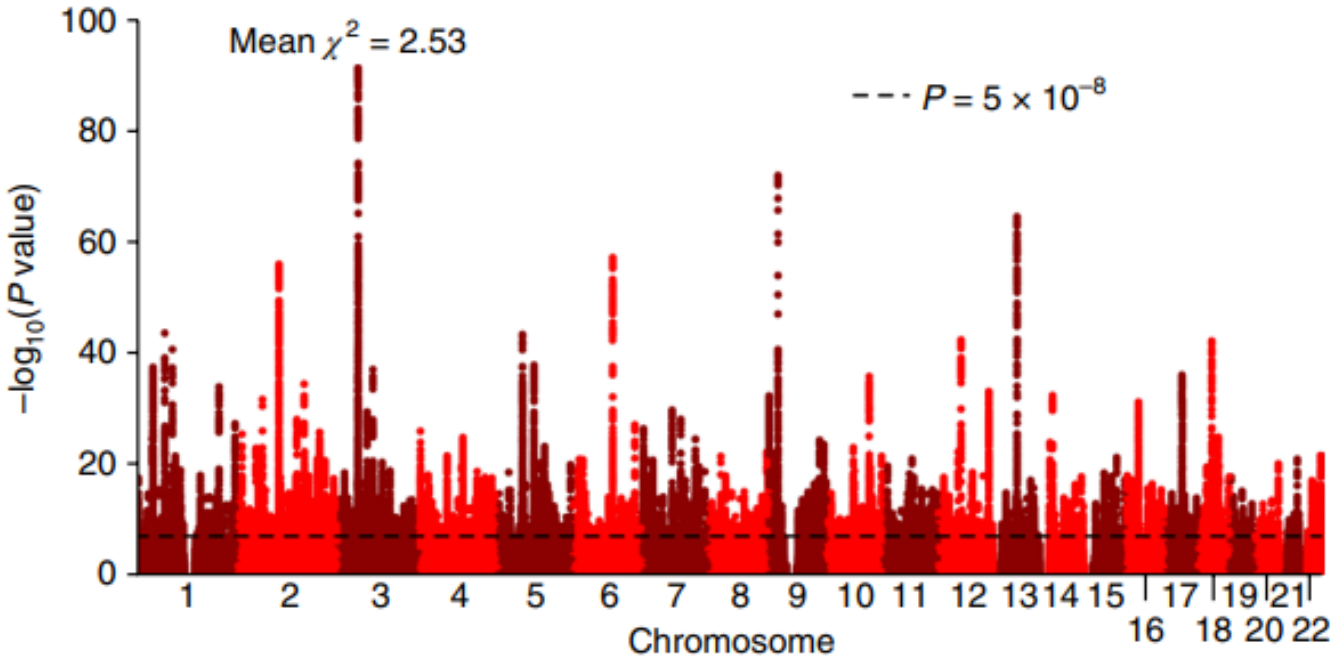


**Fig. 1 | Manhattan Plot for GWAS of EduYears.** The *P* value and mean

"A large-scale genetic association analysis of educational attainment in a sample of approximately 1.1 million individuals and identify **1,271 independent genome-wide-significant SNPs... generating a Polygenic Index** that explains 11–13% of the variance in educational attainment and 7–10% of the variance in cognitive performance." *Abstract — Lee et al. (2018, Nature Genetics)*

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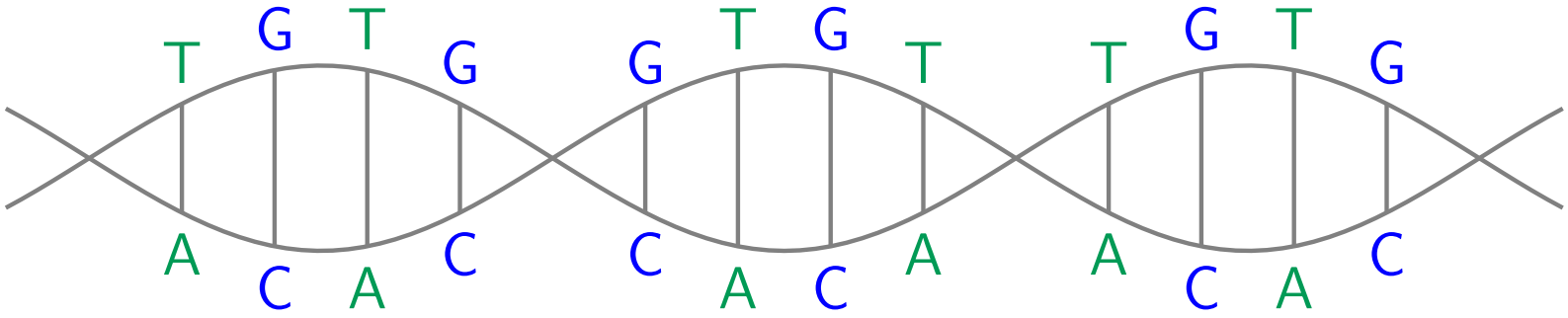


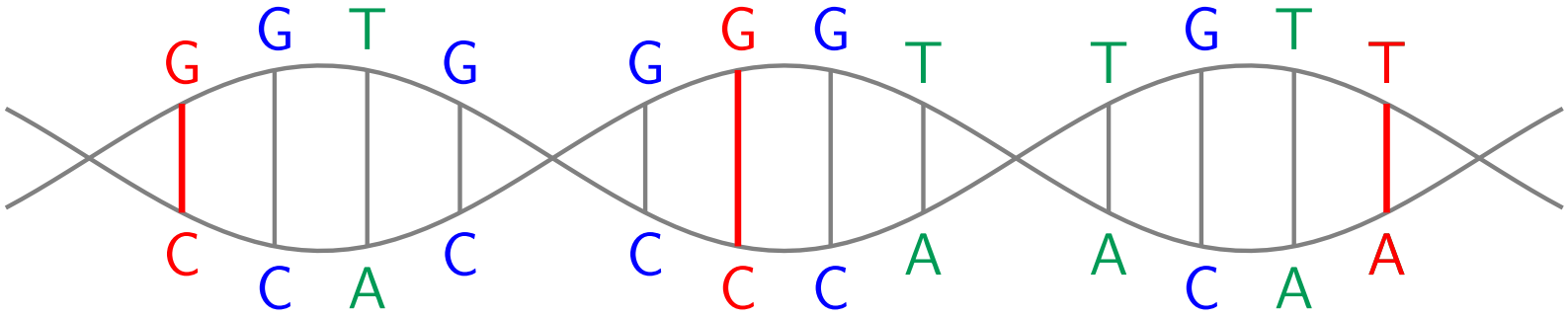

$\geq$1,271 Different Genes:

$\geq$1,271 Different Proteins: $X$

Education Attainment

# Polygenic Index & Education (Ed PGI)

Person 2's DNA:




$\geq$1,271 Different Genes:

$\downarrow$

$\geq$1,271 Different Proteins: $\boxed{X}$

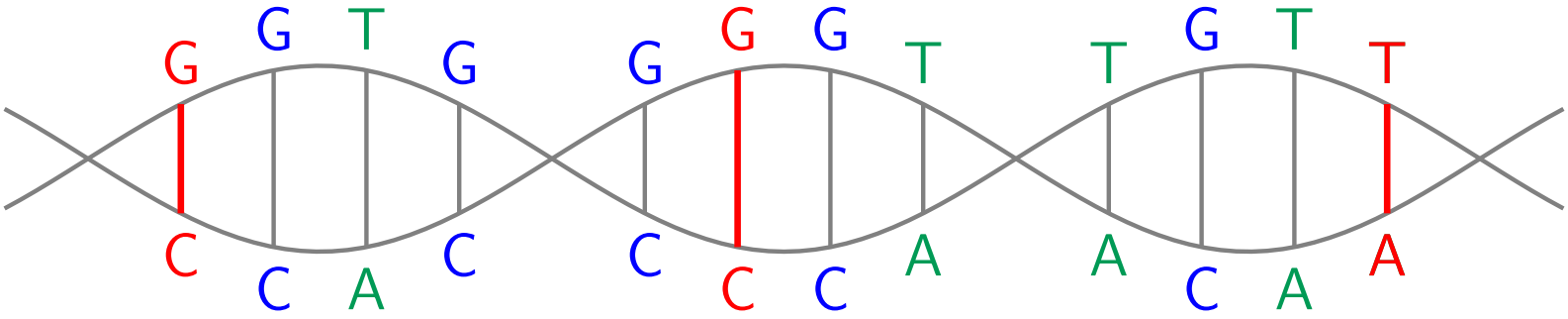
$\downarrow$

Education Attainment

- In practice, we do not know exact location of relevant gene regions
- Nor the associated protein + what they do

# Polygenic Index & Education (Ed PGI)

Person 2's DNA:




Calculate Polygenic Index (PGI):

1. Identify mutation regions (1,271 for ED PGI)
2. Calculate weights for each
3. Assign a 1-dim genetic measure, summarising + dimension-reducing this topic

Education Attainment

- In practice, we do not know exact location of relevant gene regions
- Nor the associated protein + what they do

# Polygenic Index & Education (Ed PGI)

"A large-scale genetic association analysis of educational attainment in a sample of approximately 1.1 million individuals and identify **1,271 independent genome-wide-significant SNPs... generating a Polygenic Index** that explains 11–13% of the variance in educational attainment and 7–10% of the variance in cognitive performance." *Abstract — Lee et al. (2018, Nature Genetics)*

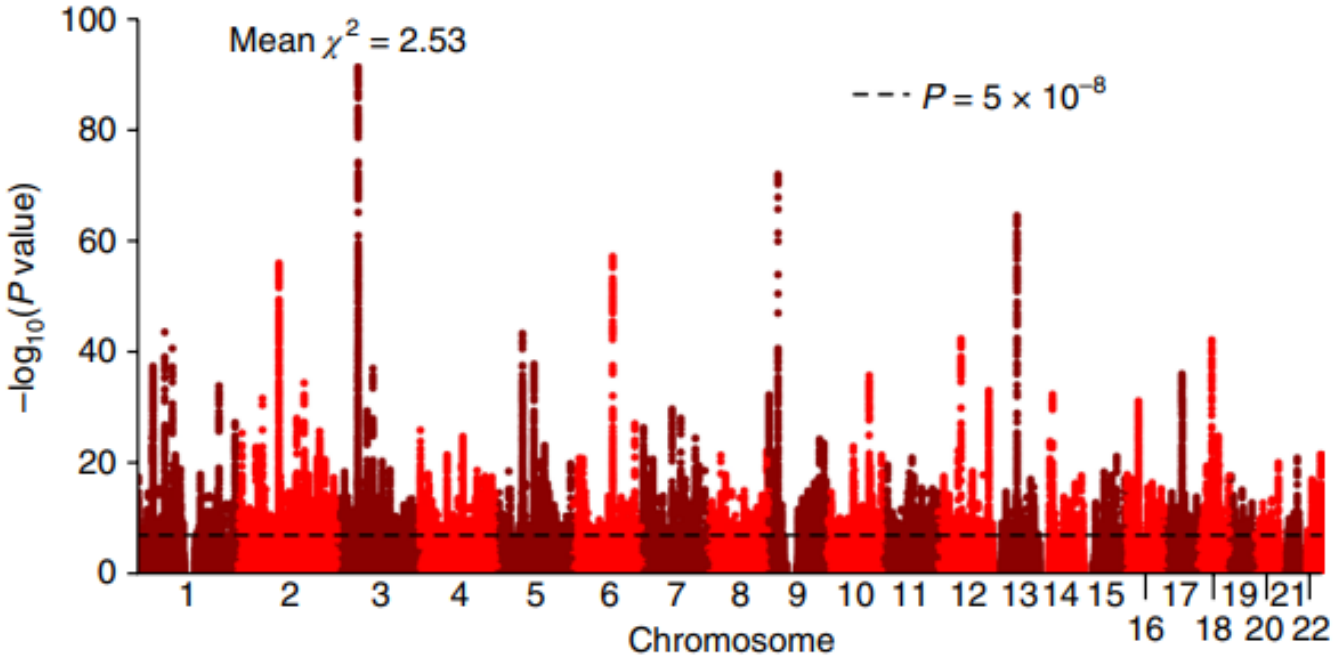
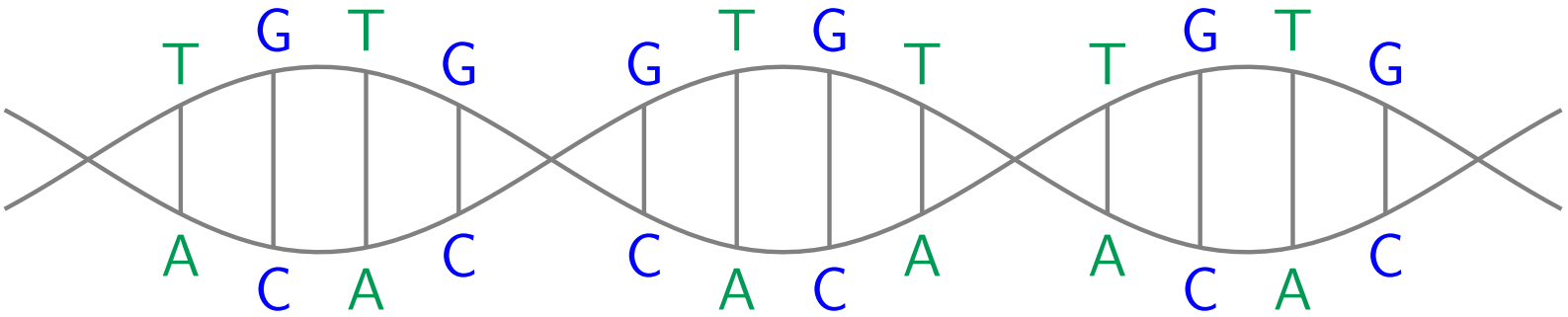



**Fig. 1 | Manhattan Plot for GWAS of EduYears.** The *P* value and mean

# Polygenic Index & Education (Ed PGI)

**Education Polygenic Index (Ed PGI):**
A weighted sum of single letter gene changes predicting years of education.

Person 1's DNA:




Weights: $w_1$ 0 0 0 0 $w_2$ 0 0 0 0 0 $w_3$

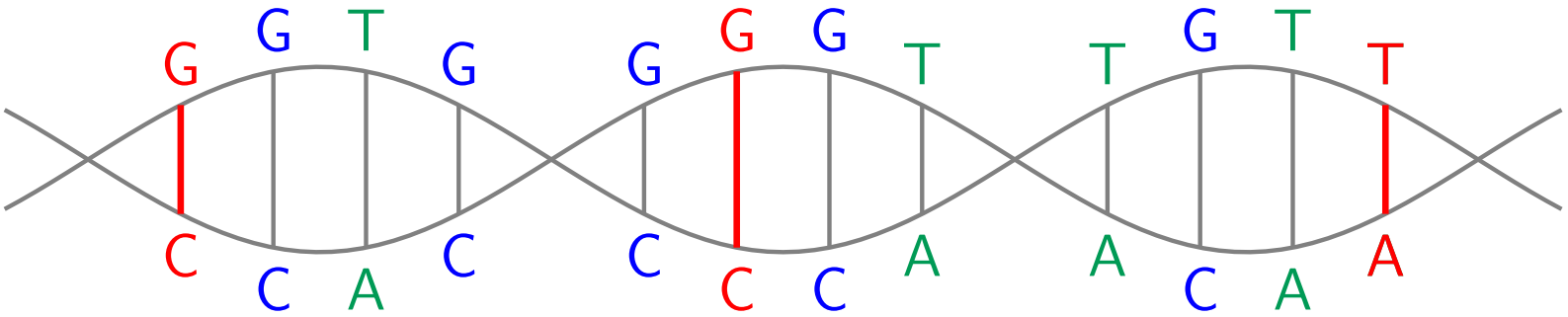
Person 1's Ed PGI:

$$Z = \sum_{j \in J} w_j \mathbb{1}\{\text{one-letter change at } j\} = 0$$

# Polygenic Index & Education (Ed PGI)

**Education Polygenic Index (Ed PGI):**
A weighted sum of single letter gene changes predicting years of education.

Person 2's DNA:




Weights: $w_1$ 0 0 0 0 $w_2$ 0 0 0 0 0 $w_3$

Person 2's Ed PGI:

$$Z = \sum_{j \in J} w_j \mathbb{1}\{\text{one-letter change at } j\} = w_1 + w_2 + w_3$$

**Note:** Humans have two copies of each chromosome, so indicator is 0, 1, 2 corresponding to how many copies one has.

# Education Polygenic Index (Ed PGI)

Previous economics + science research shows Ed PGI is strongly associated with multiple labour market outcomes:

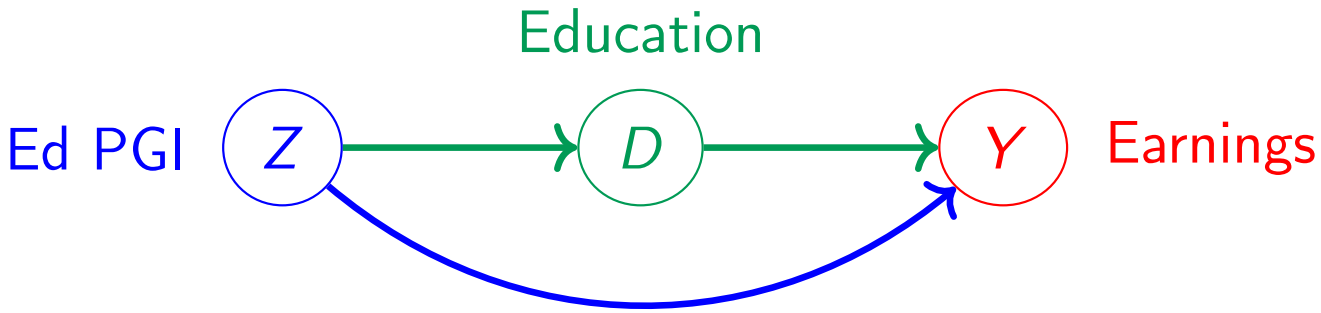
| | | |
|---|---|---|
| 1. | Retirement wealth | (Barth et al., 2020) |
| 2. | Life-cycle income | (Barth et al., 2022) |
| 3. | Longevity | (Marioni et al., 2016) |
| 4. | Primary school performance | (Aagaard-Houmark et al., 2022) |
| 5. | Parental–child care (nurture) | (Aagaard-Houmark et al., 2024) |
| 6. | Socioeconomic Status | (Carvalho, 2024) |
| 7. | ... | |
| 8. | Many in-progress | (Forthcoming) |

**Open question:** What are the mechanisms behind Ed PGI's effects?

- Is education the **only way** it is related to outcomes?
- Is there a **direct genetic relationship**, absent returns to education?

# Education Polygenic Index (Ed PGI)

I look at how Ed PGI **causally affects** labour market earnings, and **model how much this relationship is because of the education channel.**



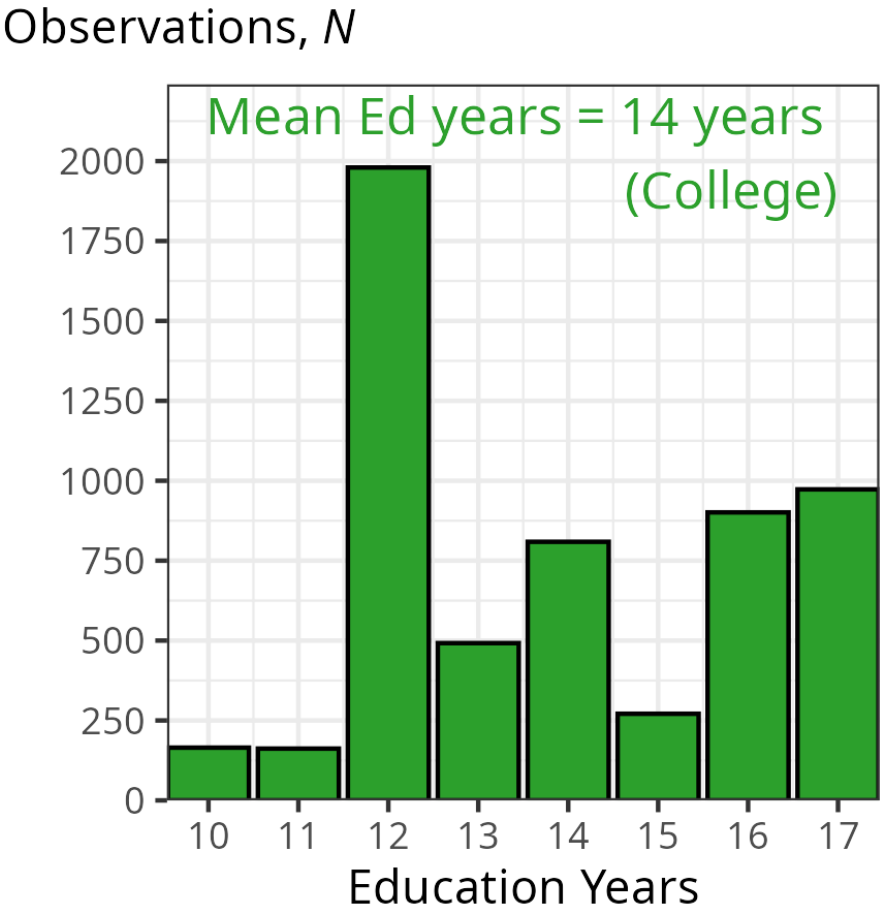

| Total Effect | = | Indirect Effect | + | Direct Effect |
|---|---|---|---|---|
| (Ed PGI → Earnings) | | (Education) | | (Genetic) |

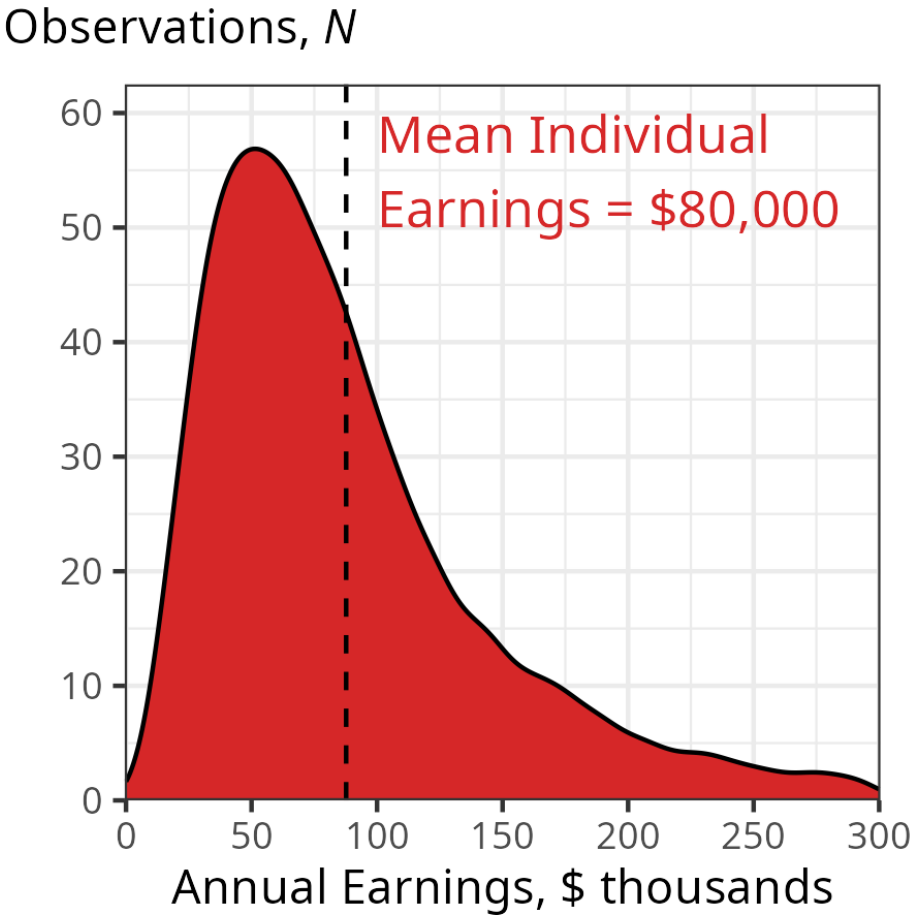
**Contributions:**

- Connect new genetic measures to classic labour econ topics
- Mediation methods in a natural experiment setting
- **Test:** *"Genetic measures correlated with education affect earnings* ***beyond just education****" — Barth Papageorge Thom (2020).*

# Data: Health & Retirement Study

A representative panel of retirement-age Americans 1990–2021.




$N = 5,957$ household heads with data on labour outcomes, age 50–65.[1]

[1] Limited to race = white, for whom Ed PGI is defined.

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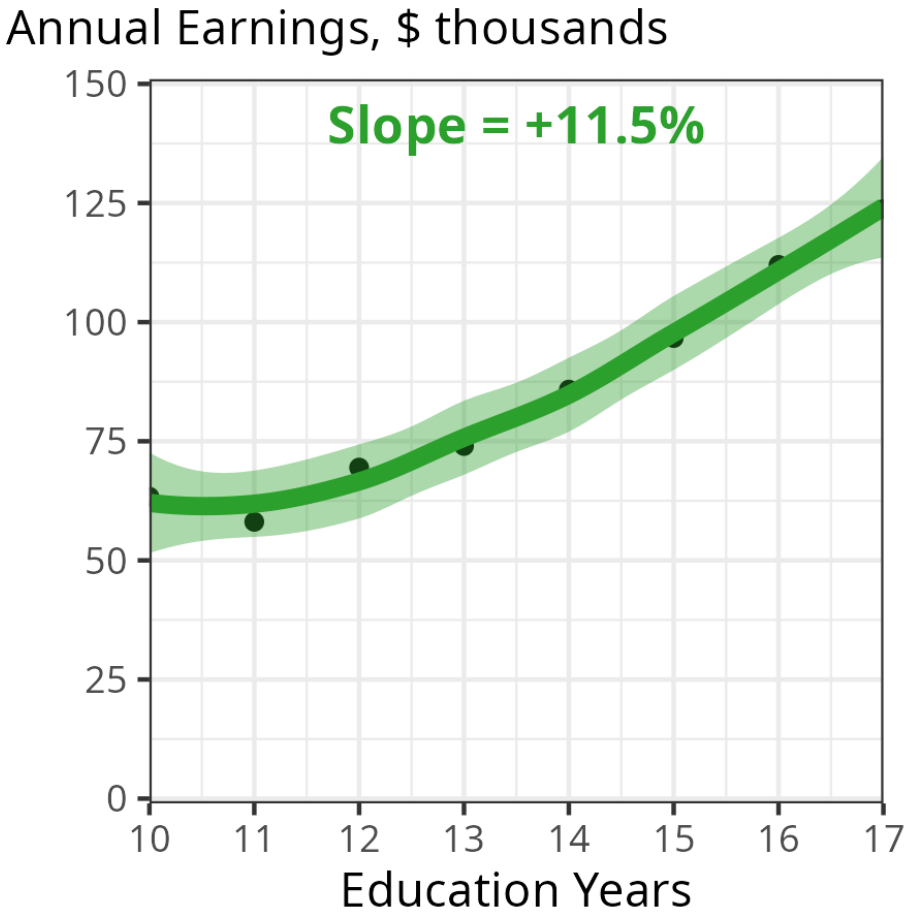


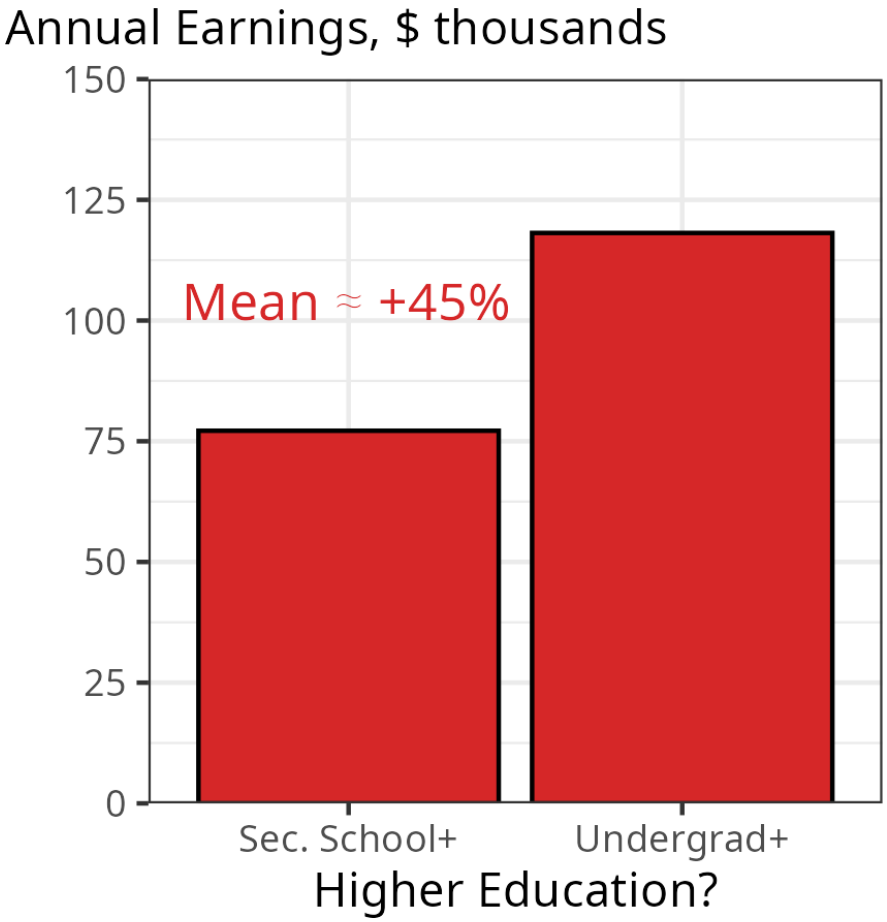


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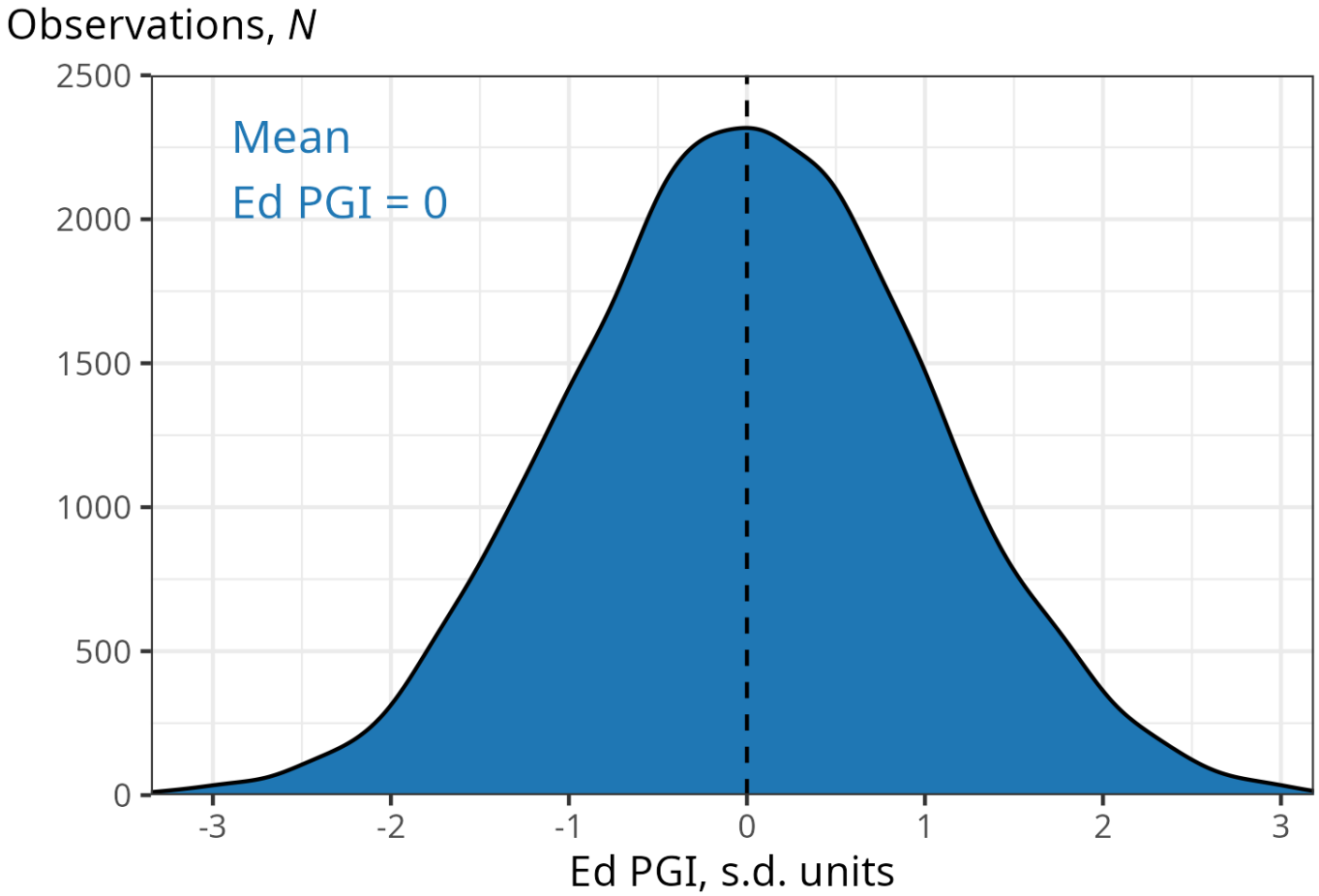


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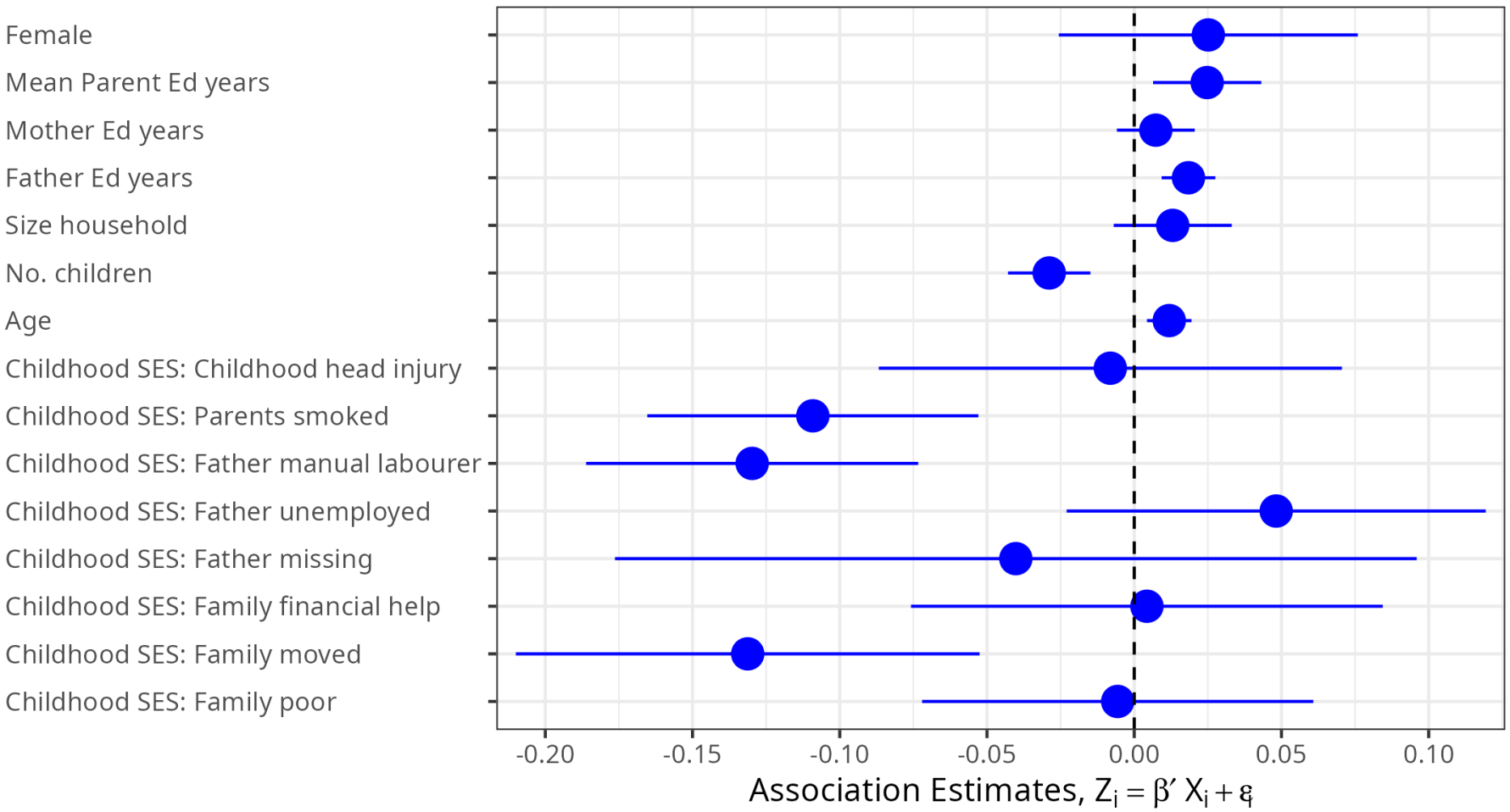
# Genetic Associations: Independent Assortment

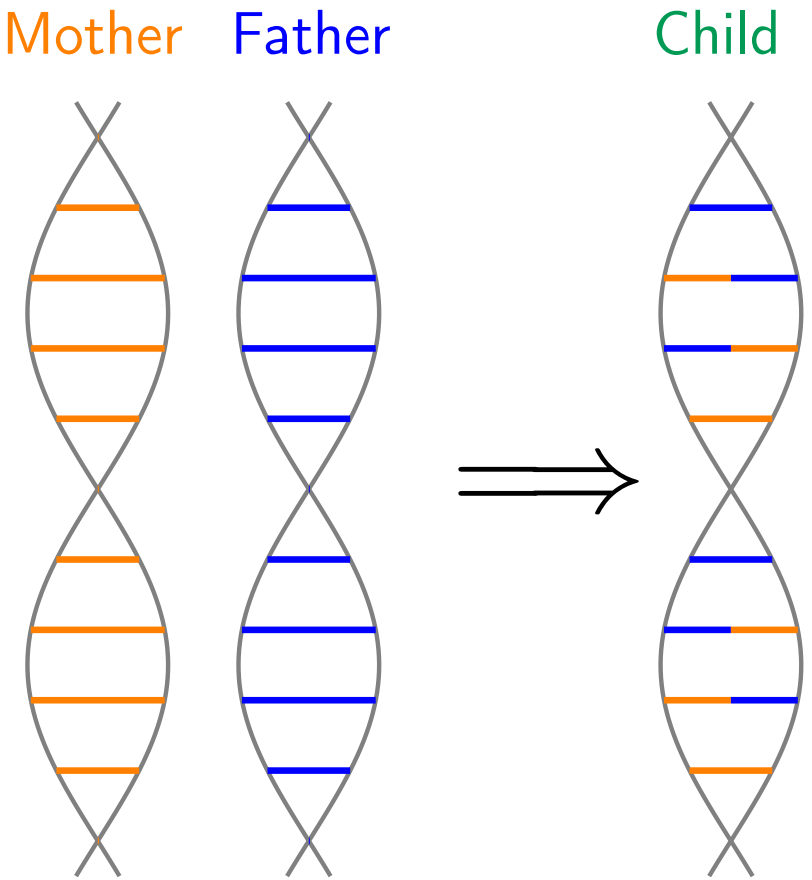
**Question:** Is Ed PGI related to other demographics?

Demographic Information



# Genetic Associations

**Question:** Is Ed PGI randomly assigned?




No, it is inherited from parents.

$$\mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_i^{\text{Mother}}, Z_i^{\text{Father}}\right]$$

$$= \frac{1}{2}\left(Z_i^{\text{Mother}} + Z_i^{\text{Father}}\right)$$

No data on parents' indices in HRS

$\Longrightarrow$ **results here not causal effects.**

# Genetic Associations: Independent Assortment

Difference in Ed PGI from parents' arises from random inheritance + mixing, Mendelian Independent Assortment.

$$Z_i^{\text{Child}} - \mathbb{E}\left[Z_i^{\text{Child}} \,\middle|\, Z_i^{\text{Mother}}, Z_i^{\text{Father}}\right] \longrightarrow \text{Independently assigned.}$$

Abbot Mendel (1886) worked through principles of genetic inheritance (Stenseth Andersson Hoekstra, 2022 PNAS).

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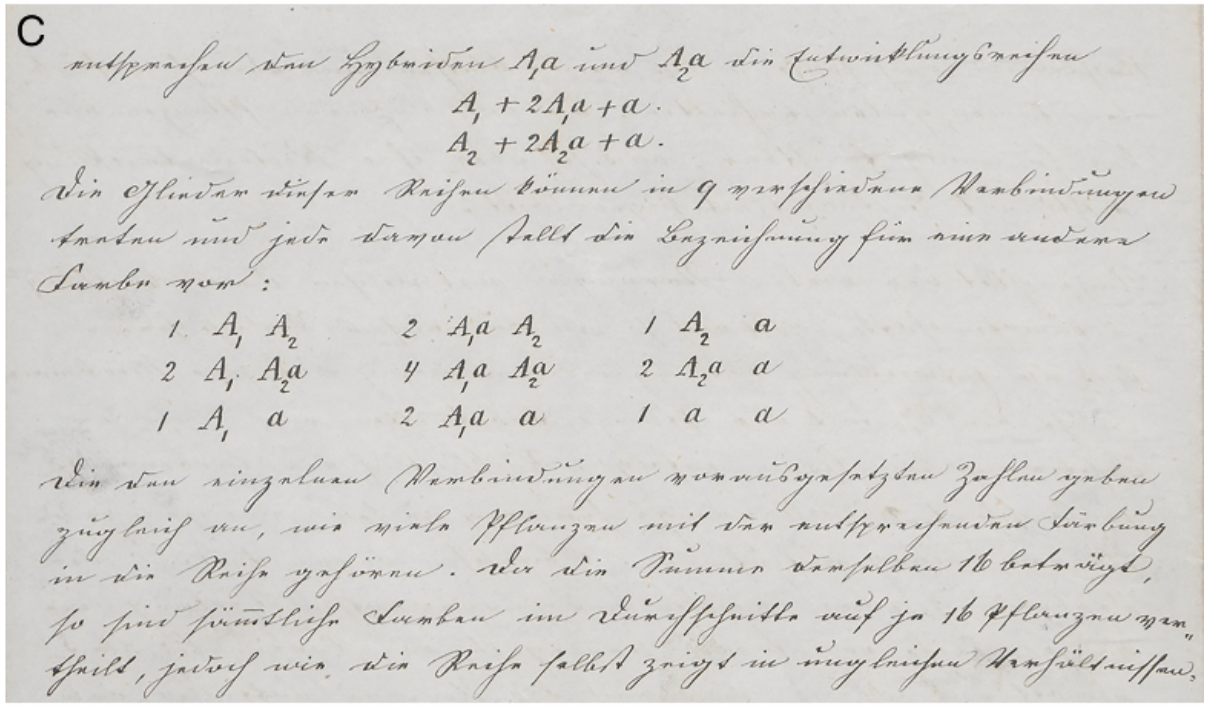
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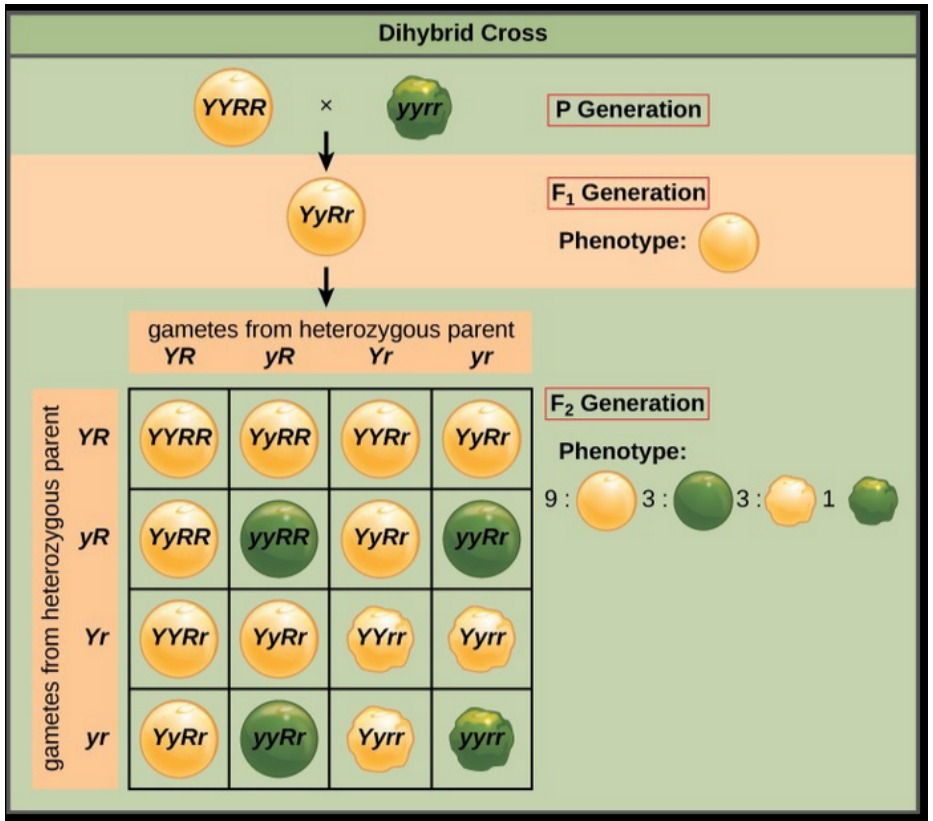


*Figure 18.9 In a dihybrid cross, two traits are followed in a single cross. Here, both seed color and seed smoothness are followed into the F2 generation.*

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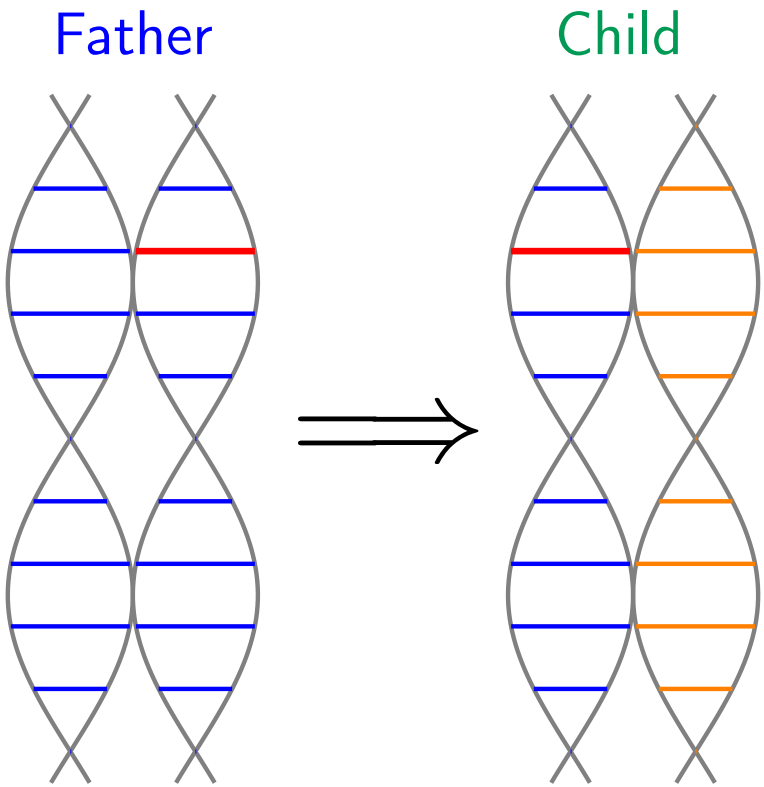


$$\Pr\left(\text{Child inherits} \,\middle|\, \text{Father one copy}\right) = \frac{1}{2}$$

Propagate this reasoning across all 1,271 single-letter changes in the Ed PGI, and both parents:

$$Z_i^{\text{Child}} - \frac{1}{2}\left(Z_i^{\text{Mother}} + Z_i^{\text{Father}}\right)$$

$$\longrightarrow \text{Independently assigned.}$$

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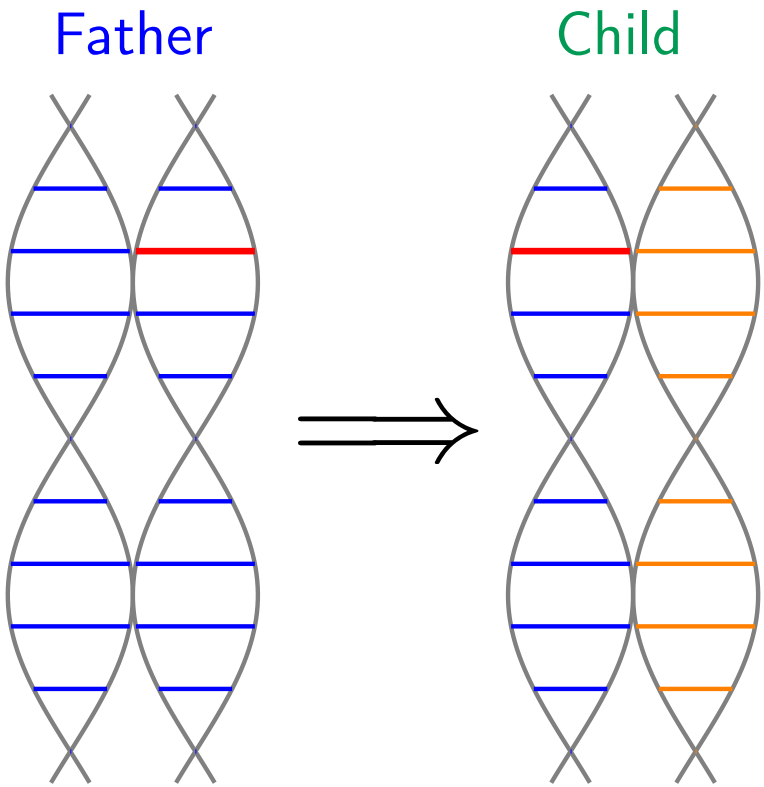



$\Longrightarrow$ Use the difference from parents as the plausibly random genetic measure.

$$Z_i^{\text{Child}} - \frac{1}{2}\left(Z_i^{\text{Mother}} + Z_i^{\text{Father}}\right)$$

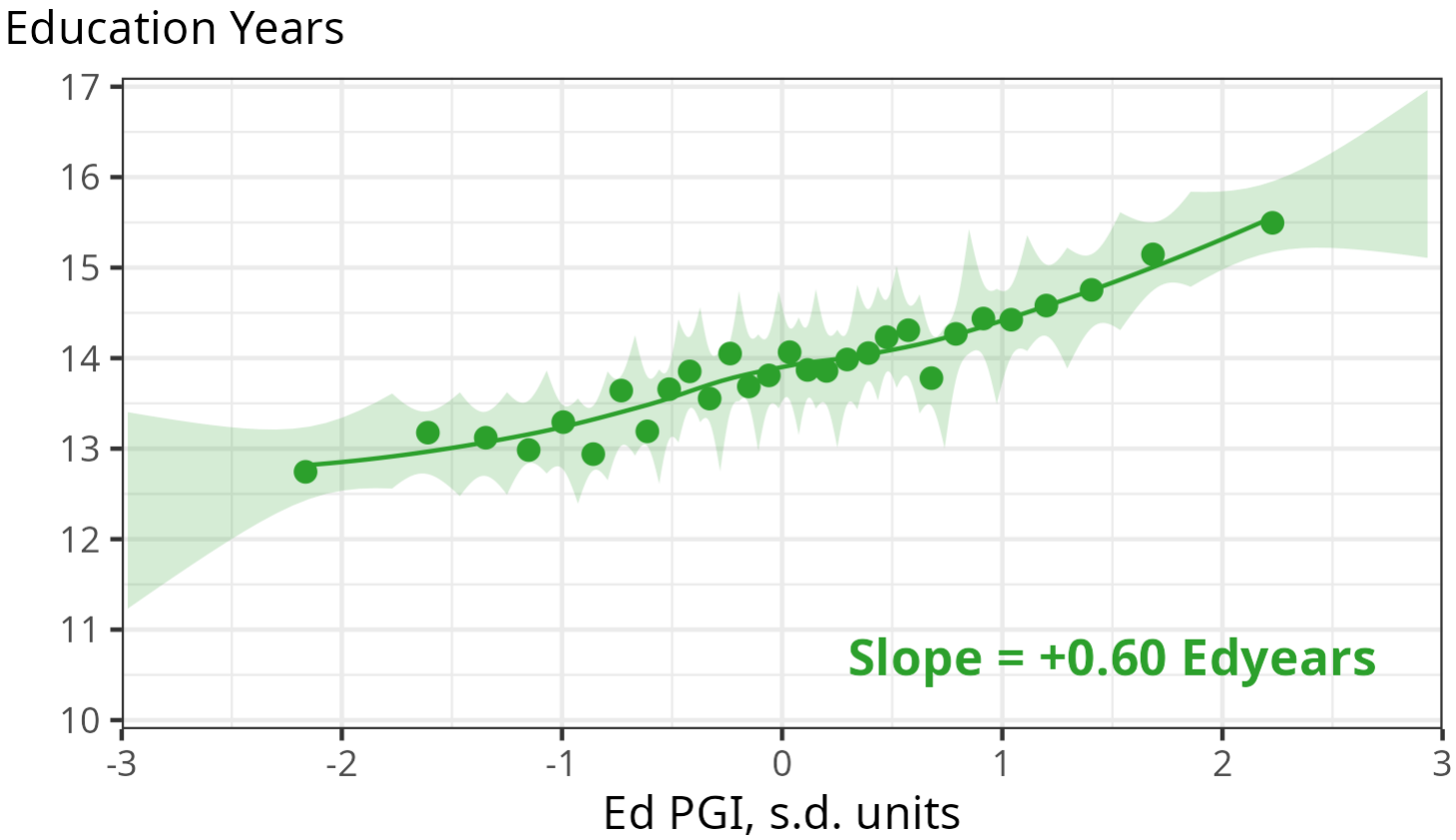
**Identification Remark:**
Equivalent to using deviation from parents as instrument for Child's Ed PGI (e.g., formula instrument).

# Genetic Associations: Outcomes

**Question:** Is Ed PGI related to labour market outcomes?

Ed PGI $Z \longrightarrow D$ Education

**Figure:** Bin Scatter Between Ed PGI + Education Years.



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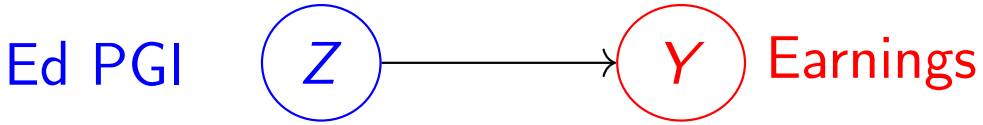
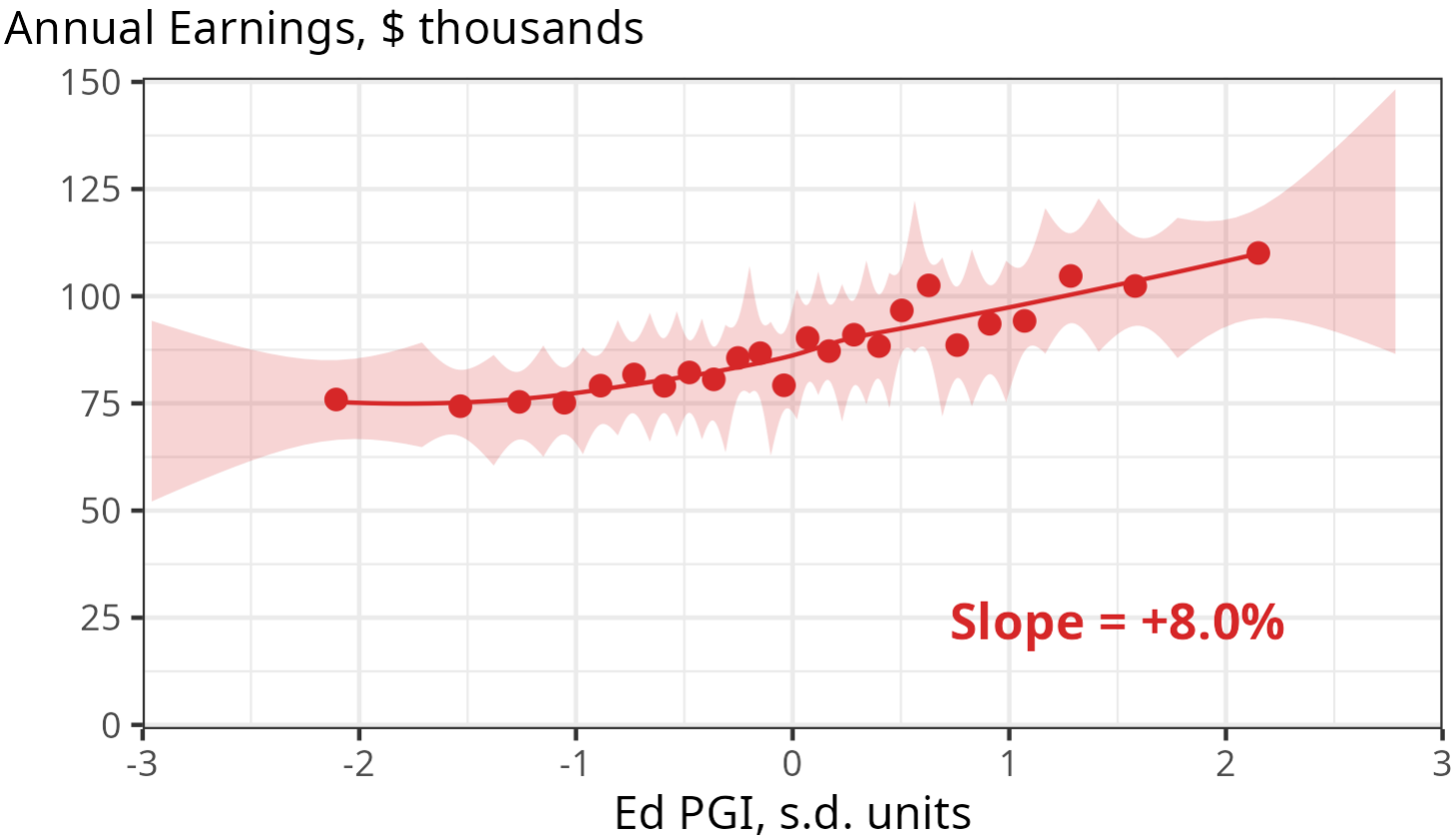
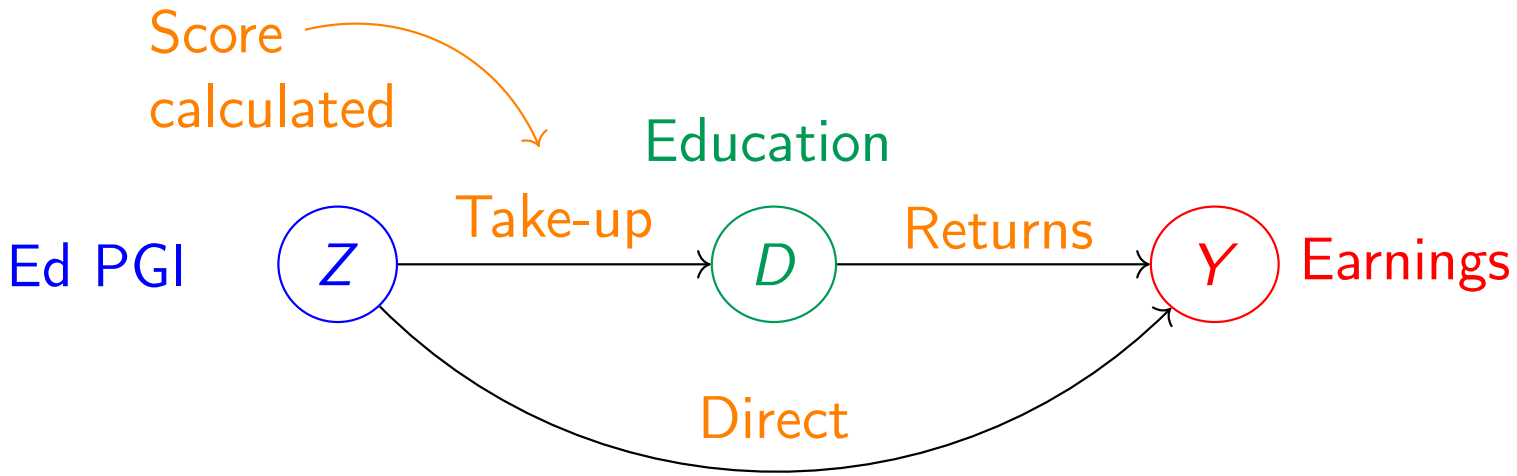



**Figure:** Bin Scatter Between Ed PGI + Annual Earnings.



# Direct and Indirect Effects

Ed PGI is correlated with **education and income**, so what?
→ Ed PGI computed from only **data on genes** + **education**.




⟹ Association with earnings is non-trivial, can operate two channels:

| Returns to Education | Direct Genetic Channel |
|---|---|
| ▶ Classroom learning | ▶ Creativity |
| ▶ Hard skills | ▶ Soft skills |
| ▶ Occupational qualifications | ▶ Signalling(?) |

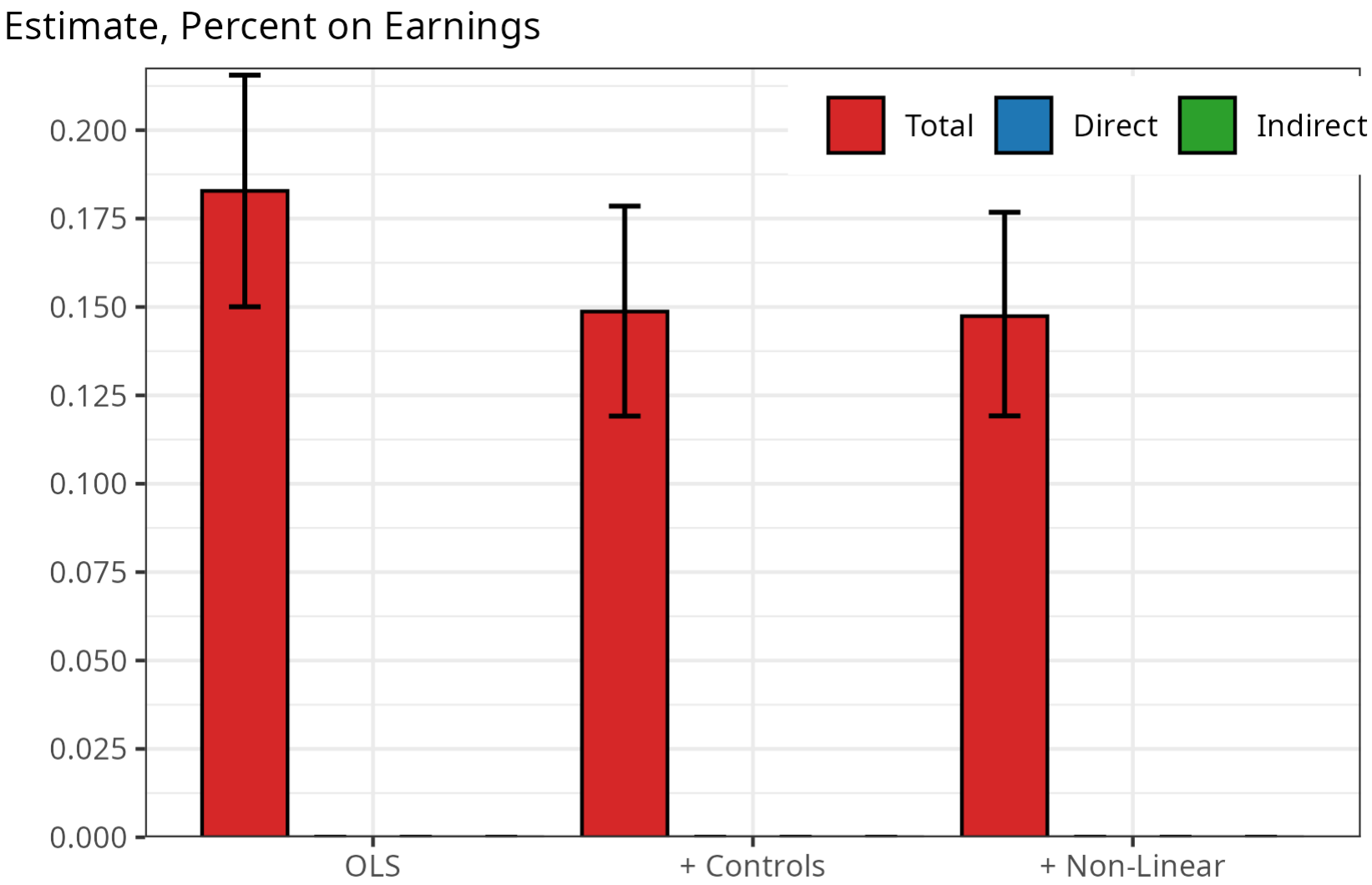
# Direct and Indirect Effects

Introduce mediation methods. General purpose decomposition of along mechanism pathways.

Equations

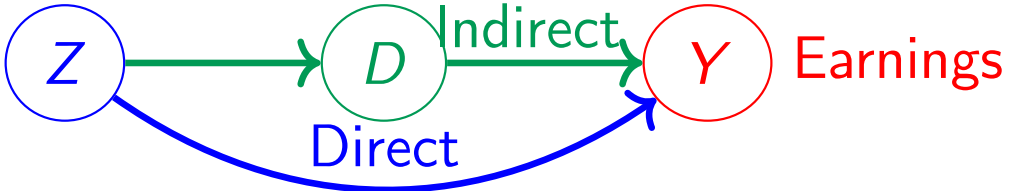
# Direct and Indirect Effects

Total Association: Ed PGI $Z \longrightarrow Y$ Earnings

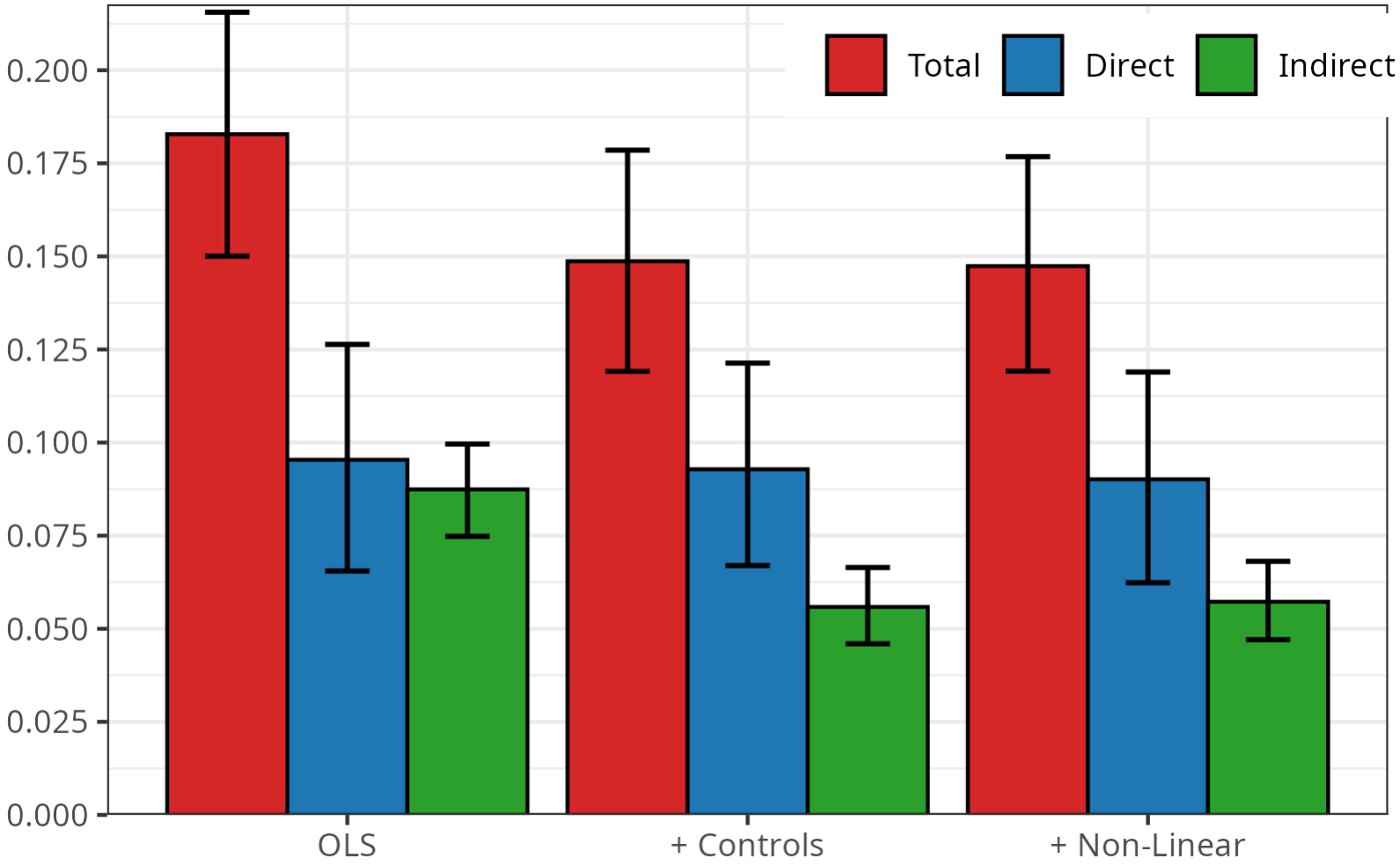


# Direct and Indirect Effects

Decomposition: Ed PGI

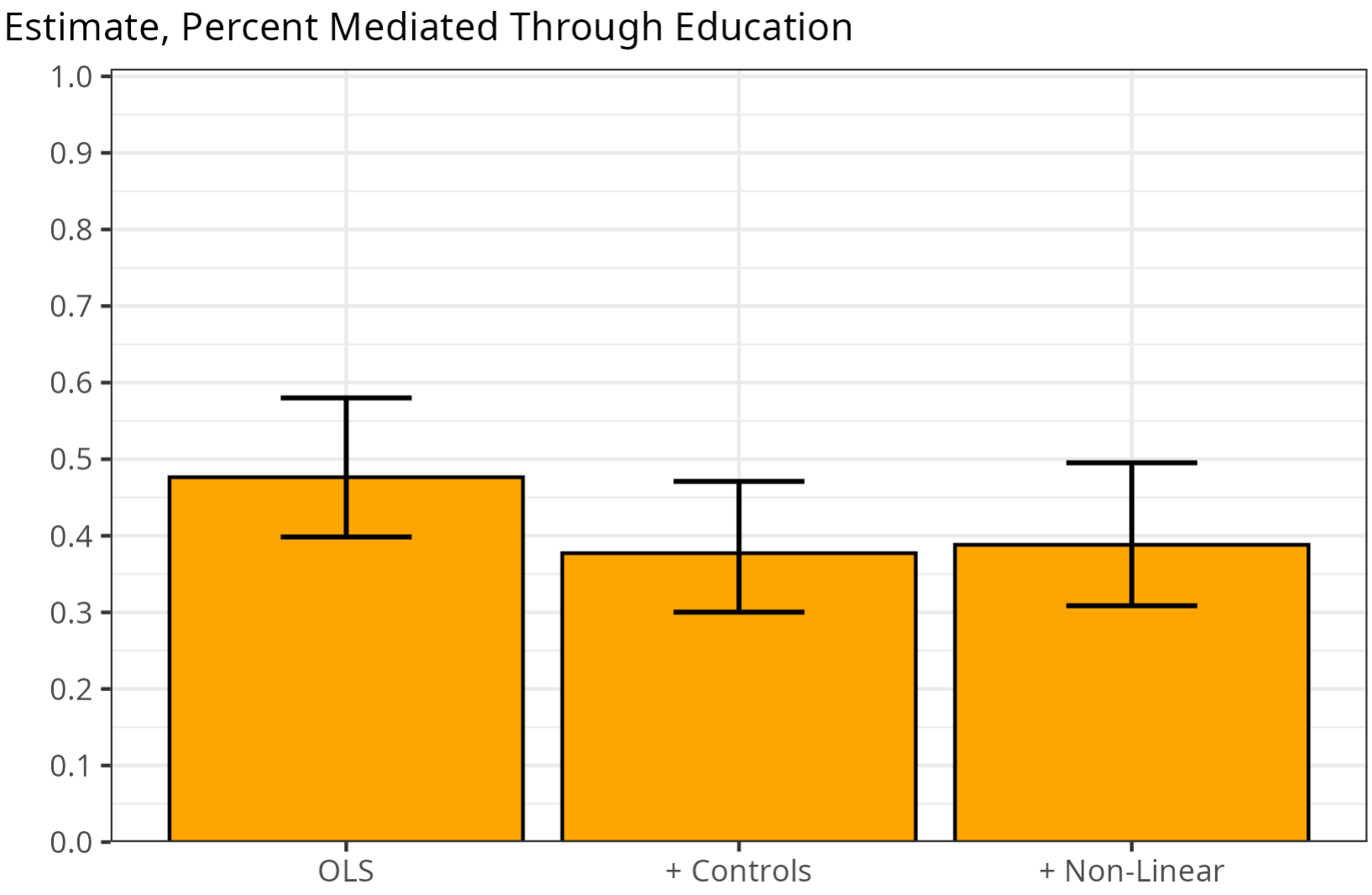



Estimate, Percent on Earnings



# Direct and Indirect Effects

Percent through education:



# Direct and Indirect Effects

Mediation methods assume Education, $D_i$, is randomly assigned (conditional on demographics) for a valid decomposition.

- Major reason why these methods not used in labour/observational research, despite popularity in other fields.
- I address this head on, by explicitly modelling selection bias on *unobserved factors* (e.g., what you want to control for **but can't**)
- Structural Roy model, selection into Education, $D_i$, based on gains.

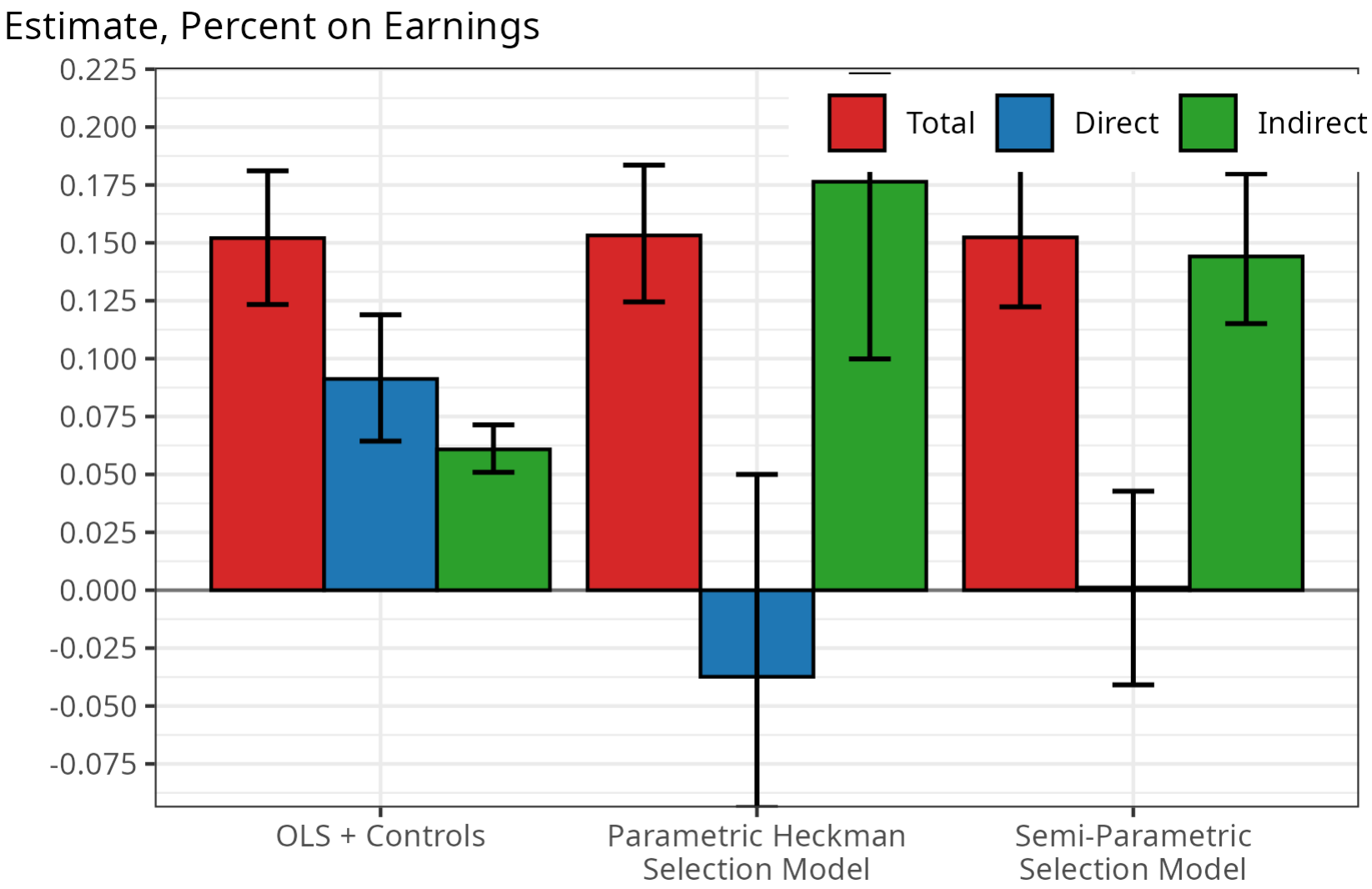
Binary choice to attend higher education:

$$D_i(Z_i) = \underbrace{\mathbb{1}\left\{\text{Educ Gains} \geq \text{Educ Costs}\right\}}_{\text{Function of demographics and unobserved factors}}$$

$\implies$ adjust estimates for unobserved factors via a selection model.
$\rightarrow$ Connects mediation to classic labour selection models, alternative solution to "assume education $D_i$ is conditionally random."

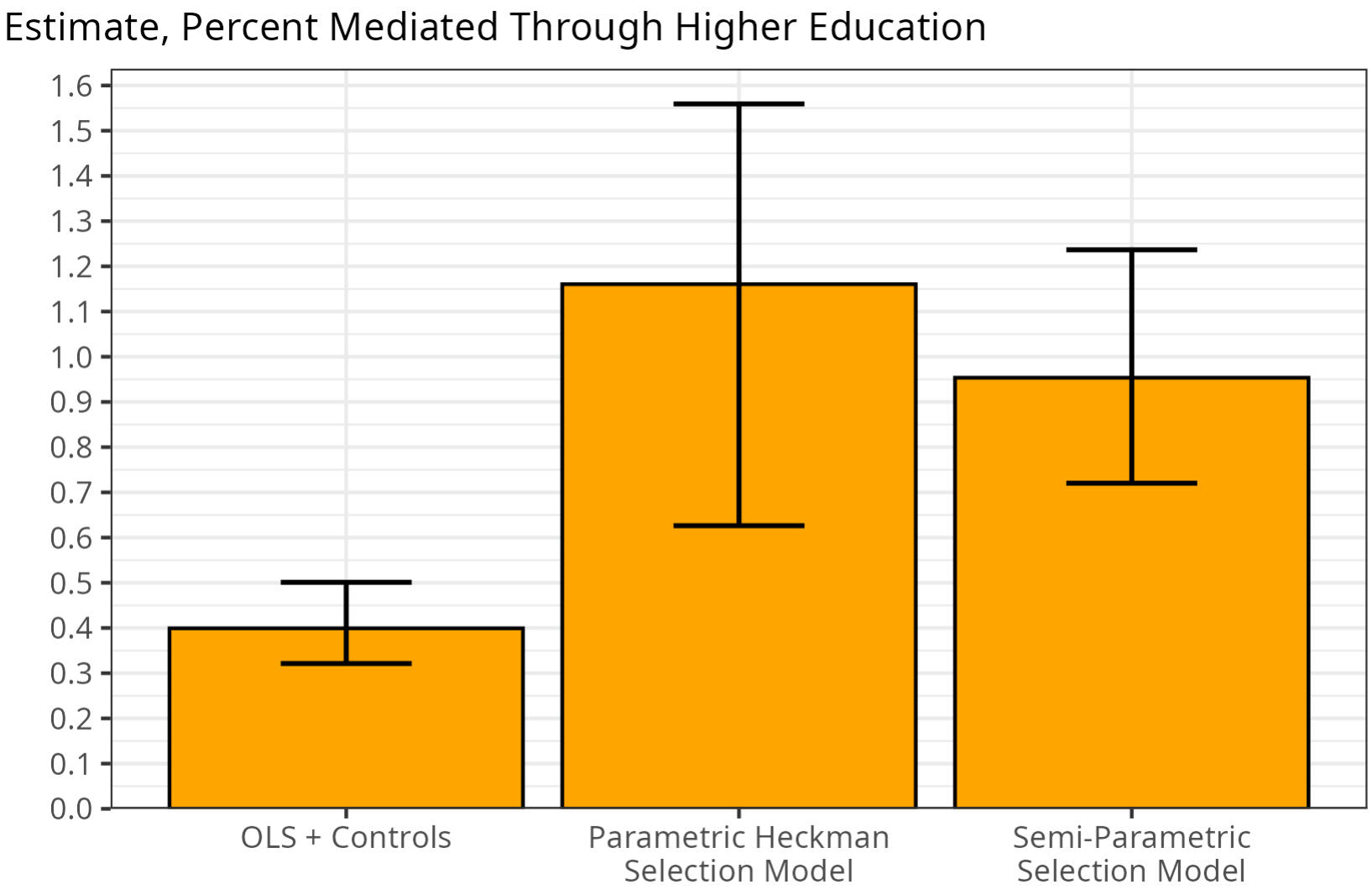
# Direct and Indirect Effects

Selection model decomposition:



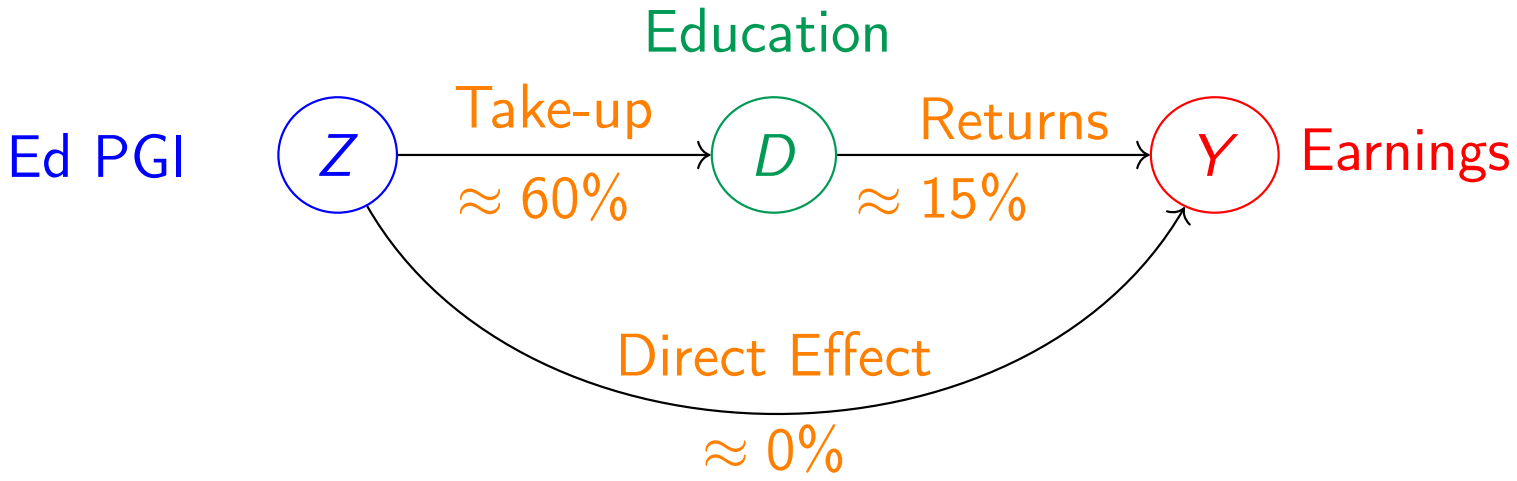
# Direct and Indirect Effects

Selection model decomposition:



# Direct and Indirect Effects

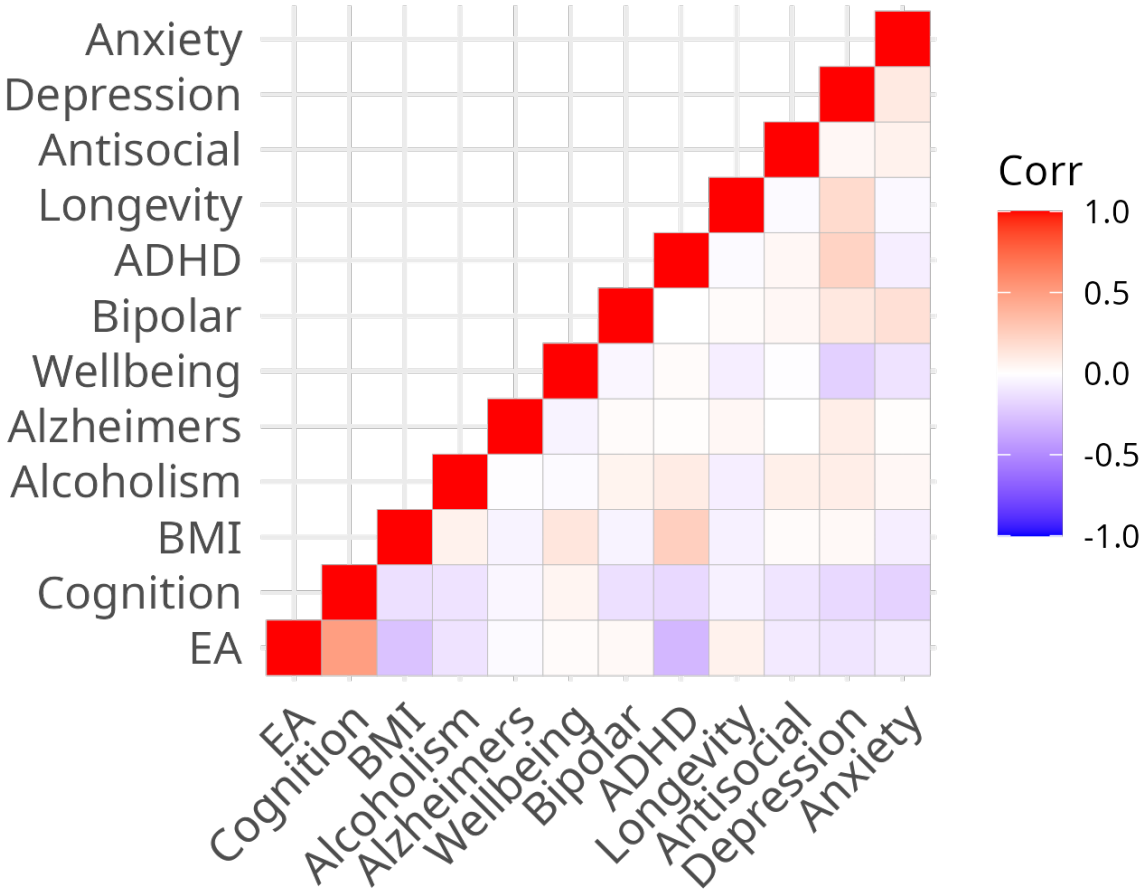
Total association Ed PGI $\rightarrow$ earnings is up to 100% education channel $\implies$ direct genetic channel $\approx$ 0%.




"The evidence suggests **fluid intelligence and self-control partly mediate** the relationship between the Ed PGI and education."
*Abstract — Carvalho (2024, JPE Micro)*

# Discussion

The claims that genetic measures (EA score) have direct effects survives the causal analysis. → **Implication: genetic predetermination**?




EA score negatively correlated with genetic predisposition to e.g., ADHD. **Implication:** education + labour market unwelcome to these factors, and

# Conclusion

Summary here

1. Studied Ed PGI associations, with education + earnings.
2. Estimated direct + indirect components of Ed PGI associations.
3. Adjusting estimates for a (structural) selection-into-treatment setting, showing **the direct channel is not distinguishable from zero, disagreeing with claims of "direct genetic effects" in contemporary research.**

Adam Smith quote here.

Thank you for your listening to my presentation.

Comments welcome now, later, or at seh325@cornell.edu.

# Appendix: Mediation Equations

Mediation methods decompose the total effect to direct + indirect channels. E.g., simultaneous regression (Imai Keele Yamamoto, 2010):[4]

$Z_i \leftarrow$ Ed PGI

$D_i \leftarrow$ Education years

$Y_i \leftarrow$ log(Earnings)

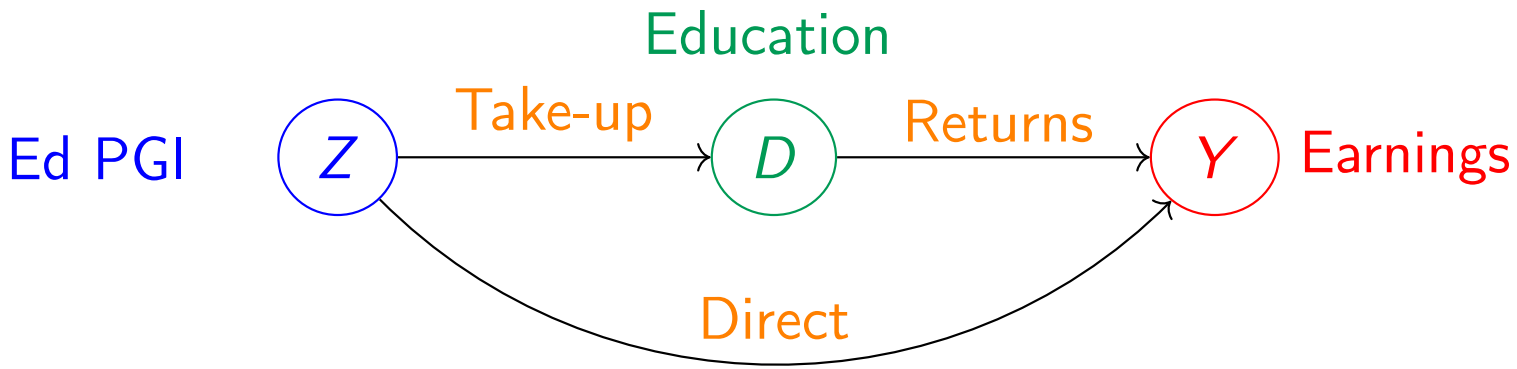
First-stage: $D_i = \theta + \pi Z_i + \eta_i$

Second-stage: $Y_i = \alpha + \beta D_i + \gamma Z_i + \delta Z_i D_i + \varepsilon_i$

$$\Longrightarrow \text{Direct} = \gamma + \delta \mathbb{E}[D_i]$$

$$\text{Indirect} = \pi(\beta + \delta \mathbb{E}[Z_i])$$

i.e., a regression decomposition of this model:




[4] Usually estimated with added controls, $+\zeta' X_i$.